

COMPARISON OF CONTINUOUS MONITOR (TEOM) AND GRAVIMETRIC SAMPLER PARTICULATE MATTER CONCENTRATIONS

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ABSTRACT. *Tapered element oscillating microbalance (TEOM) samplers may offer significant advantages to state air pollution regulatory agencies and air quality researchers in terms of reduced labor and data processing requirements through an automated particulate matter (PM) monitoring system. However, previous research has shown that TEOM samplers may not report accurate PM concentrations due to the operating characteristics of the automated system. This article presents the results of a multiyear study using collocated TEOM and gravimetric samplers configured to measure TSP concentrations from a Texas cattle feedlot. The objective of this work was to define the relationship between PM concentrations measured by TEOM and gravimetric samplers and characterize the influence of concentration intensity and particle size on that relationship. The results show that there was a significant positive linear relationship between the concentrations measured by the TEOM and gravimetric TSP samplers (p -values < 0.001). It was observed that in general, the TEOM samplers reported lower TSP concentrations than the collocated gravimetric TSP sampler. Further investigation into these results indicated that the difference in the concentration measured by the TEOM sampler versus the gravimetric TSP sampler (known as the TEOM measurement error) is correlated with the concentration measured by the gravimetric TSP sampler; but the nature of that relationship varies by location. However, linear relationships were observed between the measurement error of the TEOM samplers and the mass median diameter and geometric standard deviation of the collocated gravimetric TSP sample.*

Keywords. *Air sampler, Measurement error, Particle size distribution, TEOM, TSP.*

The particle size indicator on which particulate matter (PM) air quality regulations in the U.S. is based has changed over the years from a total suspended particulate (TSP) basis to the current PM₁₀ and PM_{2.5} basis. PM₁₀ and PM_{2.5} refer to the mass fractions of PM suspended in ambient air having nominal aerodynamic diameters less than or equal to 10 and 2.5 micrometers (μm), respectively. The National Ambient Air Quality Standards (NAAQS) for PM are ambient concentration limits established to protect public health (primary standards) and public well being (secondary standards). The original primary and secondary NAAQS for TSP was 260 $\mu\text{g m}^{-3}$ based on a 24 h average. This standard was changed in 1987 to regulate the concentration of PM₁₀.

The primary and secondary PM₁₀ NAAQS established by the EPA in 1987 were 150 $\mu\text{g m}^{-3}$ (24 h average) and 50 $\mu\text{g m}^{-3}$ (annual average) (Federal Register, 1987). In 1997, the NAAQS was further modified to include a standard for regulating PM_{2.5} (Federal Register, 1997). The primary and secondary NAAQS for PM_{2.5} established by the EPA in 1997 were 15 $\mu\text{g m}^{-3}$ (annual average) and 65 $\mu\text{g m}^{-3}$ (24 h average). Subsequent to the 2006 review of the PM NAAQS, the EPA Administrator modified the PM₁₀ standard by removing the annual average concentration limit of 50 $\mu\text{g m}^{-3}$ while retaining the 24 h average primary and secondary standard of 150 $\mu\text{g m}^{-3}$ (Federal Register, 2006). The Administrator also modified the PM_{2.5} standard by lowering the 24 h average primary and secondary standard to 35 $\mu\text{g m}^{-3}$ while maintaining the annual average concentration limit (primary and secondary) of 15 $\mu\text{g m}^{-3}$ (Federal Register, 2006).

Ambient PM_{2.5} and PM₁₀ (and formerly TSP) concentration monitoring for regulatory purposes must be conducted in accordance with federal guidelines. The EPA publishes a list of approved sampling devices that may be used to measure ambient PM concentrations for regulatory purposes (EPA, 2005). All of the samplers approved by the EPA have been designated as federal reference method (FRM) or federal equivalent method (FEM) samplers according to 40 CFR Part 53 (CFR, 1999). Work by Buser (2004) showed that FRM PM₁₀ and PM_{2.5} samplers overstate the true concentrations of PM₁₀ and PM_{2.5}, respectively, when sampling dusts with mass median diameters (MMD) larger than the cutpoint of the sampler. Agricultural operations (including feedyards, cotton gins, and grain processing facilities) emit dusts with MMDs larger than

Submitted for review in April 2007 as manuscript number SE 6970; approved for publication by the Structures & Environment Division of ASABE in December 2007. Presented at the 2005 ASABE Annual Meeting as Paper No. 054048.

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10 μm (Capareda et al., 2005). Consequently, PM_{10} and $\text{PM}_{2.5}$ samplers may overstate the true concentration of PM_{10} and $\text{PM}_{2.5}$ respectively, when sampling agricultural dusts. Field work by Capareda et al. (2005) at a Texas cotton gin indicated that PM_{10} samplers overstate true PM_{10} concentrations by 181%. A protocol described by Wanjura et al. (2004) used TSP concentration measurements along with particle size distribution (PSD) analysis to obtain more accurate measurements of PM_{10} concentrations from agricultural operations.

Included in the list of EPA-approved PM_{10} samplers is the Rupprecht & Patashnick (R&P) Series 1400a ambient PM_{10} monitor (R&P Series 1400a TEOM monitor, Rupprecht & Patashnick Co., Inc., Albany, N.Y.). This R&P monitor obtained its status as an automated equivalent method for PM_{10} in 1990 (Federal Register, 1990) and has since become a popular device used by state air pollution regulatory agencies (SAPRAs) to continuously monitor PM_{10} concentrations. When configured as designated for its PM_{10} federal equivalence method status, the R&P Series 1400a monitor uses the same PM_{10} inlet as the FRM PM_{10} sampler identified by Buser (2004) to exhibit oversampling errors.

The R&P Series 1400a monitor uses the tapered element oscillating microbalance (TEOM) method to measure PM_{10} concentrations on a real-time basis (Rupprecht and Patashnick, 2002). When the R&P Series 1400a monitor is configured as it was approved as a FEM PM_{10} sampler, it consists essentially of a sample inlet head that allows particles less than 10 μm to pass into the instrument, the TEOM sensor unit with mass transducer, and the control unit. The monitor can also be configured to measure TSP or $\text{PM}_{2.5}$ concentrations using size-selective inlets manufactured by R&P. Although the 1400a is not designated as the FRM or FEM sampler for measuring TSP or $\text{PM}_{2.5}$ in the U.S., it has been approved by the German EPA for measuring TSP and PM_{10} concentrations (Thermo Fisher, 2007). The manufacturer specifies precision of the 1400a as $\pm 1.5 \mu\text{g m}^{-3}$ (1 h average) or $\pm 0.5 \mu\text{g m}^{-3}$ (24 h average). Additionally, the mass measurement accuracy of the instrument is specified by the manufacturer as $\pm 0.75\%$ (Thermo Fisher, 2007).

Several sources have indicated that the Series 1400a monitor, when configured to measure $\text{PM}_{2.5}$ or PM_{10} concentrations, tends to report different concentrations than those measured by gravimetric FRM samplers. Hitzenger et al. (2004) found that $\text{PM}_{2.5}$ concentration measurements made by the R&P Series 1400a monitor could be as much as 18% lower than comparable gravimetric measurements. They concluded that these differences in $\text{PM}_{2.5}$ concentrations may be partially attributable to the evaporation of volatile materials in the sample stream due to the heating of the sample air stream to 50°C before entering the sensor unit to remove particle-bound water that may effect mass measurements. Similarly, Eatough et al. (2003) found that volatile particulate matter makes up a considerable portion of urban fine particulate matter ($\text{PM}_{2.5}$) and will not be measured if the sample stream is heated such that these materials volatilize and subsequently are not deposited on the sampler filter media.

The opposite result was observed by Vega et al. (2003) in a study comparing TEOM PM_{10} concentrations to U.S. FRM gravimetric PM_{10} sampler concentrations taken at five sites near Mexico City, Mexico. Over the two-month testing period, it was concluded that the TEOM concentration

measurements were consistently higher than the FRM gravimetric PM_{10} concentrations. The degree to which concentration values were overstated varied with the PM_{10} concentration level. Vega et al. (2003) indicated that statistical tests showed that the TEOM and FRM gravimetric samplers did not yield equivalent measurements and that they should not be used interchangeably.

A successful method to correct continuous beta-attenuation monitor concentrations to gravimetric PM_{10} concentrations taken every fourth day was developed by Gehrig et al. (2005). They concluded that this same method would be successful if used to correct real-time concentrations measured by TEOM monitors to concentrations measured by FRM gravimetric samplers. The implications of this work are that continuous TEOM monitors can be used with collocated gravimetric samplers that take concentration measurements on a less frequent basis (approx. every 4 d) to give accurate PM_{10} concentration monitoring results with increased time resolution. Decreasing the number of gravimetric samples taken significantly reduces monitoring costs, as gravimetric measurements are time and labor intensive to obtain.

The Center for Agricultural Air Quality Engineering and Science (CAAQES) at Texas A&M University has collected several data sets from collocated TEOM samplers (configured to measure TSP concentrations) and low-volume TSP samplers from agricultural operations. The objective of this work was to define the relationship between PM concentrations measured by the TEOM and gravimetric samplers and characterize the influence of concentration intensity and particle size on that relationship. It was hypothesized that differences in TSP concentrations measured by the TEOM sampler versus the gravimetric TSP sampler are attributable to the characteristics defining the PSD (mass median diameter and geometric standard deviation) of the sampled dust. If the relationship between TEOM and TSP concentrations could be determined, it may be possible to develop a methodology similar that used by Wanjura et al. (2004) to monitor PM_{10} and $\text{PM}_{2.5}$ concentrations on a near real-time basis.

METHODS

TEOM SAMPLER PRINCIPALS AND OPERATION

The R&P Series 1400a monitor consists of three main components (fig. 1) including:

- The sample inlet designed to allow the TSP, PM_{10} , or $\text{PM}_{2.5}$ fraction of the PM concentration in the air to pass into the instrument for concentration analysis
- The TEOM sensor unit, containing the microbalance and filter chamber.
- The system control module to monitor and record sampling flow rate data, filter mass measurements, and ambient temperature and barometric pressure measurements.

Figure 1 shows the layout of the R&P Series 1400a sampler configured to measure TSP concentrations. Initially, the 16.67 L min^{-1} sampling flow rate is drawn through the inlet head from the outside. The sampling flow rate is isokinetically split into a 3 L min^{-1} sample flow rate that is passed to the TEOM sensor unit and the remaining 13.67 L min^{-1} bypass flow (auxiliary flow) that is exhausted to the

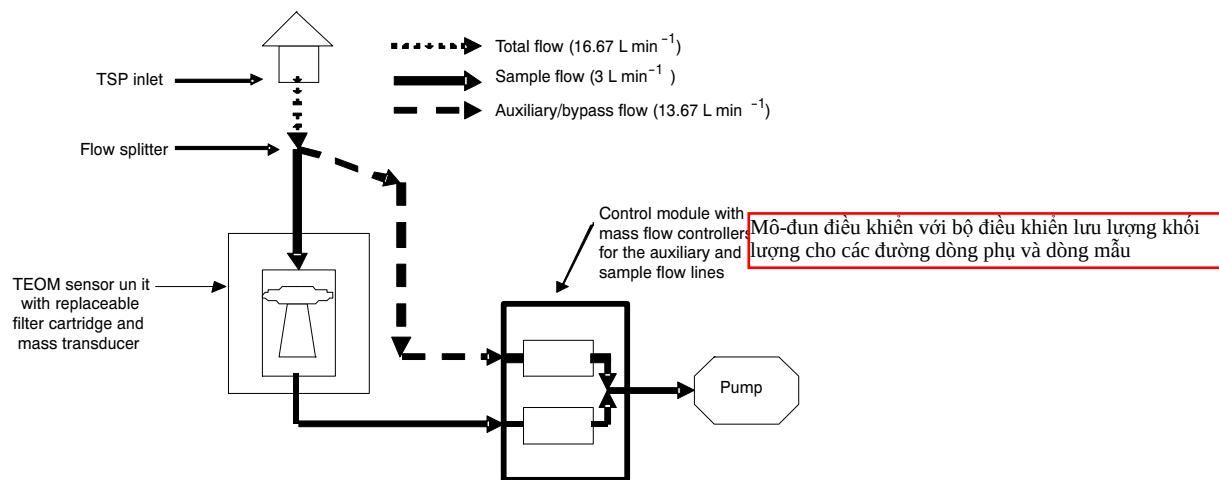


Figure 1. Schematic diagram of the R&P Series 1400a monitor configured to sample TSP concentrations.

outside. The 3 L min^{-1} sample flow is drawn through a replaceable cartridge containing a filter made of Teflon-coated borosilicate (glass fiber). Airflow through the sampler is monitored and controlled by two mass flow controllers that maintain the total flow at $16.67 \pm 1 \text{ L min}^{-1}$ and the sample flow rate at $3 \pm 0.2 \text{ L min}^{-1}$ (Rupprecht and Patashnick, 2002).

The method used by the R&P Series 1400a sampler to measure PM concentrations on a real-time basis is described by Rupprecht and Patashnick (2002).

SAMPLING METHODOLOGY

R&P Series 1400a TEOM monitors were collocated with low-volume TSP samplers (LVTSP) at a northern Texas cattle feedyard in the spring of 2004 and 2005 (fig. 2). The sampling sites located to the north and east of the feedyard were downwind sampling locations, as indicated by the

meteorological data taken during each sampling event. Based on previous sampling experience at cattle feedyards, the PSD of the PM measured downwind of the feeding pens was expected to be considerably larger than that of the upwind or background PM. In 2004, one set of collocated samplers was placed at the northern edge of the feedyard to measure feedyard emissions, and the second set was placed at the southern edge of the feedyard to measure background concentrations. The samplers were placed on the eastern and western edges of the feedyard in 2005. It was expected that the eastern sampler would measure predominately downwind concentrations from the feeding pens and from an unpaved road located between the sampling station and the feeding pens, while the western samplers would measure predominately upwind concentrations with smaller PSDs.

Equipment failures during the spring of 2005 caused data collection problems. As a result, only data from the eastern sampler were available from the sampling conducted in 2005, leaving three datasets for comparison purposes. The sampling locations from which data were available are indicated in figure 2 by the circled icons.

Each of the TEOM samplers was configured to measure TSP concentrations with a flow rate of 16.67 L min^{-1} . The TSP inlets used on the TEOM instruments were the same as those designed for the LVTSP samplers designed by CAAQES. The design of these TSP inlets was described by Wanjura et al. (2005). The sample flow rate of particle-laden air drawn through the mass transducer was 3 L min^{-1} . The remaining 13.67 L min^{-1} was exhausted to the outside of the instrument. The TEOM samplers were configured to record the total mass of PM accumulated on the filter cartridge. The mass of PM captured on the TEOM filter cartridge during each test was determined by subtracting the accumulated mass at the end of the test from the accumulated mass at the beginning of the test. The test average TSP concentration measured by the TEOM sampler was determined by dividing the mass of PM collected on the filter cartridge by the total volume of air drawn through the mass transducer.

The low-volume TSP samplers used to obtain the gravimetric TSP concentrations for this study were the same as those described by Wanjura et al. (2005). The low-volume TSP samplers used a 16.67 L min^{-1} sampling flow rate. The sampled air was drawn through a 47 mm diameter polytetrafluoroethylene (PTFE) filter ($2 \mu\text{m}$ pore size,

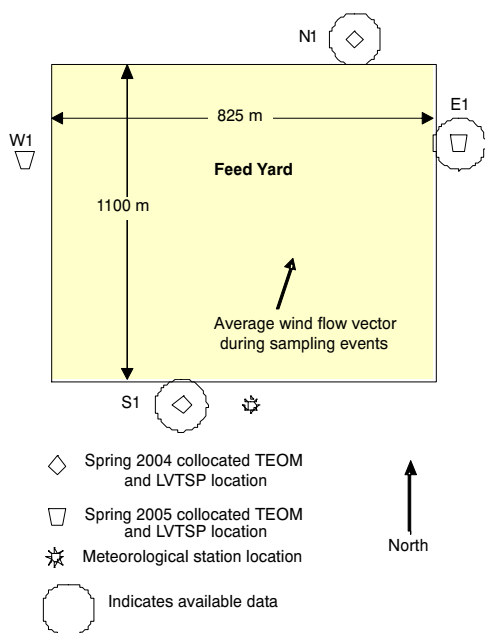


Figure 2. Feedyard schematic showing the location of the collocated TEOM and low-volume TSP (LVTSP) samplers during sampling events during the spring of 2004 and 2005. The sampling locations from which data is available are shown by the circled icons.

Zefluor membrane filter, Pall Corp., East Hills, N.Y.). All of the filters used were conditioned before and after exposure according to the protocol outlined by Wanjura et al. (2005) before being weighed on a balance (0 to 41 g range, ± 0.01 mg accuracy, AG245, Mettler Toledo, Greifensee, Switzerland). Test durations ranged from 3 to 9 h. Meteorological conditions were monitored during each test using a weather station (model S-THB-M002 temp/RH sensor and model S-BPA-CM10 barometric pressure sensor, Onset Computer Corp., Pocasset, Mass.)

Particle size distribution analyses were performed on the filters collected from the LVTSP samplers using a Coulter Multisizer3 (Beckman-Coulter, Miami, Fla.). The filter media were placed in a 5% lithium chloride methanol solution prior to being subjected to an ultrasonic bath for 5 min to remove the PM from the filter media. The Multisizer3 was configured with a 100 μm aperture tube to measure particles in the range from 2 to 60 μm . The protocol used to analyze the PSDs was described by Faulkner and Shaw (2006). Previous work by Sweeten et al. (1998) indicated that PSD analyses from filters used to measure PM concentrations from a feedyard follow the lognormal distribution relating percent mass to aerodynamic equivalent particle diameter. Thus, it was expected that the relationship between percent mass and aerodynamic equivalent diameter (AED) from the PSDs would follow a lognormal distribution described by the MMD and geometric standard deviation (GSD).

Linear regression, correlation analysis, analysis of variance (ANOVA), and one-sample t-test procedures were used to identify relationships in the measured data. The General Linear Model in SPSS (SPSS 12.0.1, SPSS, Inc., Chicago, Ill.) was used to carry out these analyses with a level of significance (α) of 0.05.

RESULTS

The concentration data collected at the S1 and N1 locations during 2004 indicate that the TEOM sampler tends to report lower concentrations than the collocated LVTSP sampler (figs. 3 and 4). However, the concentration data from the E1 location during 2005 indicate that the TEOM sampler measured concentrations both higher and lower than the collocated LVTSP sampler in equal proportions (fig. 5). Overall, the TEOM sampler tended to report lower concentrations than the LVTSP at lower concentrations

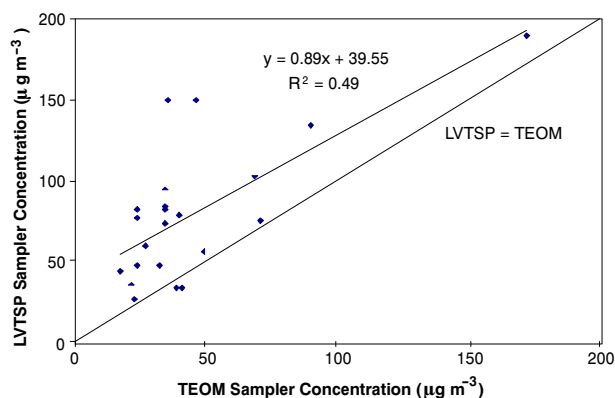


Figure 3. Plot of low-volume TSP (LVTSP) sampler concentrations versus TEOM sampler concentrations for the S1 location (see fig. 2) during the spring of 2004.

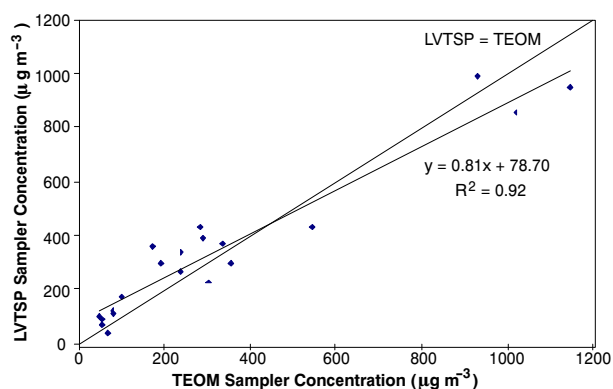


Figure 4. Plot of low-volume TSP (LVTSP) sampler concentrations versus TEOM sampler concentrations for the N1 location (see fig. 2) during the spring of 2004.

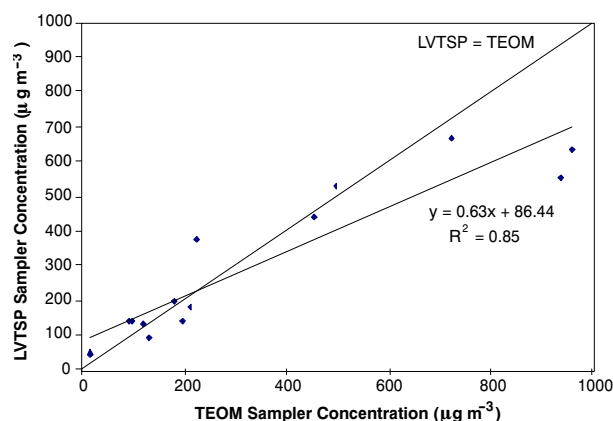


Figure 5. Plot of low-volume TSP (LVTSP) sampler concentrations versus TEOM sampler concentrations for the E1 location (see fig. 2) during the spring of 2005.

(fig. 3) and reported higher concentrations than the LVTSP at higher concentrations (figs. 4 and 5).

The results of linear regression analyses relating LVTSP sampler concentration to TEOM TSP concentration for each of the datasets are shown in table 1. As seen in figures 3 through 5 and table 1, most of the data indicated that there was a strong linear relationship (R^2 ranging from 0.85 to 0.92) between the LVTSP and TEOM sampler concentrations. However, the 2004 data from location S1 were more scattered than the other datasets and as a result had a lower R^2 value (0.49). The analysis indicated that the slopes and intercepts for the regression equations were all significant (slope p-values < 0.001 and intercept p-values < 0.03). The significant slopes are indicative of the linear relationship between LVTSP and TEOM concentration,

Table 1. Linear regression analysis results of the gravimetric vs. TEOM concentration data taken during the spring of 2004 and 2005. For each case, the measured TEOM concentration is multiplied by the slope and added to the intercept to determine the gravimetric concentration.

Dataset	Location	R^2	Regression Equation Coefficients		Intercept p-Value	Slope p-Value
			Constant	Slope		
Spring 2004	S1	0.49	39.55	0.89	0.002	< 0.001
	N1	0.92	78.70	0.81	0.006	< 0.001
Spring 2005	E1	0.85	86.44	0.63	0.029	< 0.001

while the strength of the linear relationship is indicated by the R^2 value.

The trends observed in the concentration data indicate that the concentrations measured by the TEOM sampler may be influenced by both the magnitude of the TSP concentration and the PSD of the sampled PM. The measurement error of the TEOM sampler, as defined in equation 1, was introduced to help investigate the influence of TSP concentration and PSD characteristics on the performance of the TEOM sampler:

$$E_M = C_G - C_T \quad (1)$$

where

E_M = measurement error of the TEOM sampler

C_G = concentration measured by the gravimetric LVTSP sampler

C_T = concentration measured by the TEOM sampler.

One-sample t-tests were conducted in SPSS on the TEOM measurement error values from the three datasets ($H_0: E_{M,i} = 0$ vs. $H_a: E_{M,i} \neq 0$) to determine statistical differences in the mean TSP concentrations measured by the LVTSP and TEOM samplers. A significant difference between the mean TSP concentrations was observed between the LVTSP and TEOM measurements for the 2004-S1 data (p-value < 0.0001) but not for the 2004-N1 (p-value = 0.425) or 2005-E1 data (p-value = 0.301). Although no statistical difference was observed for the 2004-N1 or 2005-E1 data, the confidence intervals indicate a substantial range on the TEOM measurement error data for these datasets. The wide confidence intervals further indicate high levels of uncertainty in the TSP concentrations measured by the TEOM samplers. The results of this analysis and the 95% confidence intervals are shown in table 2.

Regression analyses indicated that significant linear relationships exist between the measurement error of the TEOM samplers and the gravimetric TSP concentrations for the 2004-S1 (slope p-value = 0.002) and 2005-E1 (slope p-value = 0.049) datasets. However, the analysis indicated that there was no significant linear relationship for the 2004-N1 dataset (slope p-value = 0.099). None of the regression equation intercepts were significant. Combining all of the datasets, a significant indirect linear relationship was observed between the measurement error of the TEOM sampler and the LVTSP concentration (slope = -0.172, slope p-value = 0.002, intercept = 49.57, intercept p-value = 0.005). The results of these analyses are presented in table 3.

Although significant linear relationships were observed between the TEOM measurement error and LVTSP concentration for the 2004-S1, 2005-E1, and combined data, the R^2 values of these relationships are quite low. This indicates that the TEOM concentration is not influenced by the TSP concentration alone.

Table 2. Results of t-test to determine differences in the mean concentrations measured by the TEOM and LVTSP samplers as well as 95% confidence intervals (CI) on the measurement error of the TEOM samplers for the 2004 and 2005 datasets.

Dataset	Location	t-test ($H_0: E_M = 0$)		95% CI	
		t	p-Value	Lower	Upper
Spring 2004	S1	5.271	<0.0001	21.07	48.53
	N1	0.815	0.425	-28.31	64.43
Spring 2005	E1	-1.078	0.301	-124.46	41.6

Table 3. Linear regression analysis results relating the measurement error of the TEOM samplers to the measured LVTSP sampler concentration. In each case the LVTSP sampler concentration is multiplied by the slope and added to the intercept to determine the expected measurement error.

Dataset	Location	R^2	Regression Equation Coefficients		Intercept p-Value	Slope p-Value
			Constant	Slope		
Spring 2004	S1	0.398	-1.39	-.455	0.903	0.002
	N1	0.144	64.26	-0.134	0.074	0.099
Spring 2005	E1	0.284	66.14	-0.352	0.291	0.049
All data combined		0.17	49.57	-0.172	0.005	0.002

Correlation analyses performed on the TEOM measurement error, TEOM concentration, gravimetric TSP concentration, test average relative humidity, and test average temperature indicated insignificant correlations between the variables. Thus, the ambient temperature and relative humidity were observed to have little effect on the measurement error of the TEOM sampler. The results of the correlation analyses are shown in tables 4 and 5.

Linear regression analyses were performed on the data from the N1 sampling location (spring 2004) and the E1 sampling location (spring 2005) relating MMD and measurement error as well as GSD and measurement error. PSD data were not available from the 2004-S1 dataset due to the small mass of PM collected on the filters. The results of the regression analysis indicate that there is a significant linear relationship between the measurement error of the TEOM sampler and the MMD of the collocated LVTSP sampler for the 2004-N1 data (slope p-value = 0.006) and the 2005-E1 data (slope p-value = 0.009). The R^2 values of the relationship between TEOM measurement error and LVTSP MMD for the 2004 and 2005 data were 0.54 and 0.51, respectively, indicating a stronger relationship than was

Table 4. Correlation analysis results on the TEOM measurement error, TEOM concentration, and gravimetric TSP concentration data from 2004 on the test average ambient temperature and relative humidity (RH).

Variable	Temp.		RH	
	R	p-Value	R	p-Value
Location N1				
TEOM measurement error	-0.126	0.597	0.22	0.35
TEOM concentration	0.33	0.155	-0.257	0.275
Gravimetric conc.	0.211	0.323	-0.238	0.262
Location S1				
TEOM measurement error	0.091	0.687	0.054	0.811
TEOM concentration	0.143	0.514	0.019	0.93
Gravimetric TSP conc.	0.159	0.468	-0.196	0.37

Table 5. Correlation analysis results on the TEOM measurement error, TEOM concentration, and gravimetric TSP concentration data from 2005 on the test average ambient temperature and relative humidity (RH).

Variable	Temp.		RH	
	R	p-Value	R	p-Value
Location E1				
TEOM measurement error	-0.376	0.185	0.472	0.089
TEOM concentration	0.375	0.186	-0.469	0.091
Gravimetric TSP conc.	0.116	0.67	-0.229	0.393

observed between the TEOM measurement error and TSP concentration magnitude.

A significant linear relationship was observed between TEOM measurement error and LVTSP GSD for the 2005-E1 data but not for the 2004-N1 data. The R^2 values for the 2005 and 2004 data were 0.51 and 0.06, respectively. The results of these analyses are shown in tables 6 and 7.

Multiple regression analyses were performed on the 2004-N1 and 2005-E1 datasets to determine the portion of the variance in the TEOM measurement error that can be explained by the LVTSP concentration, MMD, and GSD. The R^2 values for the regression models including all of the predictor variables (LVTSP concentration, MMD, and GSD) were 0.61 and 0.67 for the 2004-N1 and 2005-E1 datasets, respectively. However, the p-values for the variable coefficients in each model indicate that some of the variables in each model are not significant (table 8) and do not significantly improve the predictive value of the relationship.

It was expected that the relationship between TSP measurements made by collocated TEOM and LVTSP samplers would follow a pattern similar to that seen by Vega et al. (2003). The results presented by Vega et al. (2003) indicated that the TEOM sampler would report higher concentrations of PM_{10} than a collocated gravimetric PM_{10} sampler. These samplers were sampling in an urban environment near Mexico City, Mexico. The EPA (1996) characterizes the PSD typical of urban environments as having an MMD of approximately 6 μm with a GSD of

approximately 2.25. The PSD characteristics of urban environments are considerably different from those of a rural environment near a cattle feedyard with typical MMD and GSD values in the range of 18 μm and 2, respectively.

The results presented here (for TSP concentration sampling) indicate that, in general, the TEOM sampler measured lower concentrations than a collocated gravimetric TSP sampler. At first glance, these results seem to counter the trends presented by Vega et al. (2003). However, combining the 2004-N1 and 2005-E1 datasets (fig. 6), linear regression analysis relating the measurement error of the TEOM sampler and the MMD of the LVTSP sampler indicates that for MMDs lower than 20 μm , the measurement error of the TEOM sampler (as defined by equation 1) is negative. The converse is true for MMDs over 20 μm . A negative measurement error indicates that the TEOM sampler is reporting higher concentrations than the gravimetric sampler. The data used to substantiate this finding exhibit considerable spread in the data points with MMD <20 μm . Thus, the strength of the relationship shown in figure 6 may be improved with further research.

Several factors could have lead to these results. For example, the operating characteristics of the TSP inlets allow a much broader range of particle diameters to enter the sample stream of the TEOM than PM_{10} inlets. In general, this causes the percentage of smaller particles (particle diameter $\leq 10 \mu m$) captured on the filter to be lower for a TSP sample than the percentage of smaller particles for a PM_{10} sample. As a result, the isokinetic sampling point in the flow splitter of the TEOM may influence the sample flow such that more of the smaller diameter particles are pulled into the sample stream that is sent to the TEOM sensor unit. This possibility is indicated in the relationship between the TEOM measurement error and the MMD of the LVTSP sampler (fig. 6). As the MMD increases, the measurement error of the TEOM moves from a negative value (indicating that the TEOM concentrations are higher than the gravimetric TSP sampler concentrations) to a positive value when the MMD is larger than 20 μm .

In addition, the deposition of particles on the TEOM filter may influence the mass concentrations in some way. If the PM mass is deposited on the filter in a concentrated location, then the physical parameters of the TEOM mass-spring system could be influenced such that the K_0 (system spring rate constant) value changes. However, this is highly

Table 6. Linear regression results relating the measurement error of the TEOM sampler to the MMD of the collocated gravimetric TSP sampler from the spring of 2004 and 2005. The independent variable in each case is the MMD of the gravimetric sample with the measurement error of the TEOM being the dependent variable.

Dataset	Location	R^2	Regression Equation Coefficients		Intercept p-Value	Slope p-Value
			Intercept	Slope		
Spring 2004	N1	0.54	-435.51	21.33	0.007	0.006
Spring 2005	E1	0.51	-618.15	32.56	0.007	0.009

Table 7. Linear regression results relating the measurement error of the TEOM sampler to the GSD of the collocated gravimetric TSP sampler from the spring of 2004 and 2005. The independent variable in each case is the GSD of the gravimetric sample. The dependent variable is the TEOM sampler measurement error.

Dataset	Location	R^2	Regression Equation Coefficients		Intercept p-Value	Slope p-Value
			Intercept	Slope		
Spring 2004	N1	0.06	-331.97	155.7	0.45	0.44
Spring 2005	E1	0.51	-867.7	363.23	0.007	0.009

Table 8. Multiple regression results to predict TEOM measurement error as a function of LVTSP concentration, MMD, and GSD for the 2004 and 2005 datasets.

	2004-N1 Model ($R^2 = 0.61$)		2005-E1 Model ($R^2 = 0.67$)	
	Coefficient	p-Value	Coefficient	p-Value
Intercept	-133.22	0.715	-787.29	0.034
LVTSP conc.	-0.1	0.287	-0.15	0.496
LVTSP MMD	20.38	0.024	12.85	0.435
LVTSP GSD	-110.83	0.515	251.13	0.09

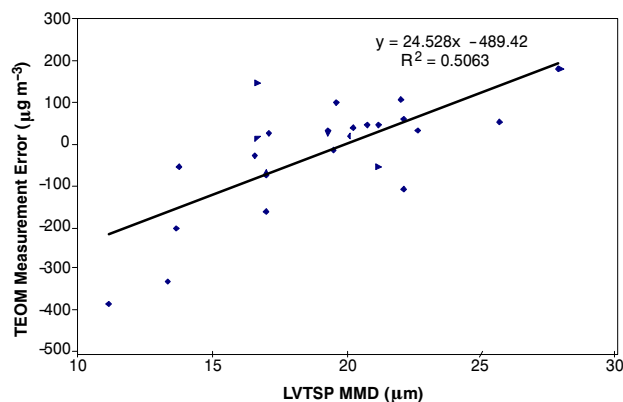


Figure 6. TEOM measurement error versus LVTSP MMD for the combined 2004-N1 and 2005-E1 datasets.

unlikely, as the TEOM units have been repeatedly shown to have a linear decrease in oscillation frequency with increasing system mass.

These factors are not presented as definitive answers to explain the differences in the concentrations measured by the TEOM samplers compared to that of the gravimetric TSP sampler, but only perhaps as suggestions for future work in this area.

CONCLUSIONS

The major findings of the work presented in this article are as follows:

There was a significant positive linear correlation between TSP concentrations measured by collocated TEOM and gravimetric samplers. In general, the TEOM sampler measured lower concentrations than the collocated gravimetric TSP sampler. Wide confidence intervals in the measurement error of the TEOM sampler for each dataset indicate large uncertainty in the concentrations measured by the TEOM samplers. Thus, accurate measurements of TSP concentrations cannot be determined by TEOM samplers alone. It is necessary to collocate gravimetric TSP and TEOM samplers in order to develop a relationship between measured and true TSP concentrations. Once established, this relationship can be used to estimate true concentrations from concentrations measured by the TEOM sampler.

The relationship between TEOM measurement error and LVTSP concentration varied by sampling location. For the upwind sampling location (2004-S1), the relationship was linear with a significant positive slope. However, the slope of the linear relationship observed for the downwind locations (2004-N1 and 2005-E1) was negative and only significant for the 2005 dataset. Combining all data, the relationship between TEOM measurement error and LVTSP concentration was observed to be indirect and significant. More investigation into the influence of concentration intensity on TEOM sampler performance is necessary.

The measurement error of the TEOM had a significant positive correlation with the MMD of the collocated gravimetric TSP sampler. Similarly, there was a positive correlation between TEOM measurement error and GSD, but it was only significant for the 2005-E1 data.

There was a stronger positive linear relationship between the measurement error of the TEOM sampler and the MMD of the collocated LVTSP sampler than with GSD.

There was no significant correlation between the concentration measured by the TEOM sampler, gravimetric TSP sampler, or the measurement error of the TEOM sampler with the ambient temperature or relative humidity during the tests. Thus, the ambient temperature and moisture had little effect on the measurement error of the TEOM sampler.

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